

Session 4:

Problem Set 4

Course Supervisor: Dr. Salem

Teaching Assistant: Zoleikani

October 25, 2025

Problem 4.1

It is assumed that the data presented below represent the estimation output of the regression model of Y on X .

$$\sum X_i Y_i = 492 \quad n = 16 \quad \bar{X} = 6 \quad \bar{Y} = 4 \quad \sum X_i^2 = 657 \quad \sum Y_i^2 = 526$$

Additional distributional critical values are provided:

$$t_{0.025,14} = 2.14 \quad \chi_{0.025,14} = 26.12 \quad \chi_{0.975,14} = 5.63492$$

Required:

- (a) Estimate the sample regression line.
- (b) Compute the variance of the slope estimator β_1 .
- (c) Construct a 95% confidence interval for the population slope β_1 .
- (d) Construct a 95% confidence interval for the error variance σ^2 .
- (e) Test the significance of the slope at the 5% level.
- (f) Compute and interpret the coefficient of determination R^2 .

Problem 4.2

Consider the classical linear regression model:

$$Y = X\beta + \varepsilon$$

where $E(\varepsilon) = 0$ and $Var(\varepsilon) = \sigma^2 I$.

Required:

- (a) Derive the OLS estimator of β .
- (b) Show that the estimator is unbiased.
- (c) Derive the variance–covariance matrix of the estimator.

Problem 4.3

Consider the model:

$$Y_i = \beta + \varepsilon_i$$

Required:

- (a) Obtain the OLS estimator of β .
- (b) Show that the estimator is unbiased and derive its variance.

Problem 4.4

You are given summary statistics from the simple linear regression of Y on X :

$$\sum X_i Y_i = 830102 \quad n = 22 \quad \bar{X} = 416.5 \quad \bar{Y} = 56.65 \quad \sum X_i^2 = 3919654 \quad RSS = 130.6$$

Critical values supplied:

$$t_{0.025, 20} = 2.09 \quad \chi_{0.025, 20}^2 = 34.17 \quad \chi_{0.975, 20}^2 = 9.59$$

Required:

- (a) Estimate the sample regression line $\hat{Y} = b_0 + b_1 X$.
- (b) Compute the variance of the slope estimator $\text{Var}(\hat{b}_1)$.
- (c) Construct a 95% confidence interval for the population slope β_1 .
- (d) Construct a 95% confidence interval for the error variance σ^2 .
- (e) Test the significance of the slope at the 5% level.
- (f) Compute the coefficient of determination R^2 and interpret it.